Psychological Research (2025) 89:121
https://doi.org/10.1007/s00426-025-02156-7

RESEARCH

Seeing the fours before the threes: investigating numerical signatures with hierarchical navon stimuli

Valter Prpic[1,2] · Arianna Felisatti[3] · David Aagten-Murphy[4] · Luisa Lugli[1] · Martin H. Fischer[4]

Received: 16 August 2024 / Accepted: 25 June 2025 / Published online: 8 July 2025

Abstract

When two numbers are compared, numerical processing signatures include the size effect (small numbers processed faster than large ones), the Spatial-Numerical Association of Response Codes (SNARC) effect (small numbers classified faster with left side than right side responses), and the distance effect (numbers farther from a reference processed faster than those closer). These signatures reflect single digit processing and their relevance for multi-digit processing relies on evidence from horizontal digit strings. In hierarchical displays, such as Navon stimuli, different information coexists at various spatial scales (local and global). Consequently, a single stimulus supports different processes depending on the attended scale. We examined number processing at two different spatial scales to investigate the influence of both the task relevant and irrelevant digits on processing signatures. We used hierarchical stimuli with symbolic numbers (1, 4, 6, 9) at global and local levels, such as a large "global" digit (e.g., 9) composed of smaller "local" digits (e.g., 1). Separate groups of participants classified either the global (N=31) or local (N=30) number magnitude relative to 5, ignoring the other scale. Consistent with Navon effects, we found faster processing for global than local stimuli and when global and local information matched. Both groups demonstrated significant SNARC and distance effects, but only the local group showed size effects. Notably, local numbers were classified faster even when the global number required the same response. Overall, these results indicate that numerical processing operates concurrently at different spatial scales, though local information is particularly vulnerable to interference from global context. This underscores the complex interplay between global and local processing in numerical cognition.

Introduction

Numbers are fundamental to everyday activities, from organizing schedules to conducting transactions, and they serve as crucial tools for measurement in engineering and scientific disciplines. For cognitive scientists, they offer a compelling lens through which to explore the intricacies of the human mind and our numerical processing capabilities. Studies on isolated number judgements often reveal consistent patterns, such as faster classification for smaller than larger numbers (*size effect;* Moyer & Landauer, 1967), faster responses for numbers far than near from the comparison value (*distance effect;* Moyer & Landauer, 1967), and faster left responses for small numbers and right responses for larger numbers (Spatial-Numerical Association of Response Codes (SNARC) *effect*; Dehaene et al., 1993). These benchmark findings have informed our cognitive models of the human mind (e.g., Dehaene & Cohen, 1995; Sixtus et al., 2023), focussing on the notion of a mental number line that represents numbers ordered by their magnitude (for a debate

✉ Valter Prpic
valter.prpic@unibo.it

✉ Luisa Lugli
l.lugli@unibo.it

✉ Martin H. Fischer
martinf@uni-potsdam.de

1 Department of Philosophy, University of Bologna, Bologna, Italy

2 Department of Theoretical and Applied Sciences, eCampus University, Novedrate, Italy

3 Department of General Psychology, University of Padova, Padova, Italy

4 Division of Cognitive Sciences, Department of Psychology, University of Potsdam, Potsdam, Germany

on the role of order and magnitude in the SNARC effect see Pitt & Casasanto, 2022 and Prpic et al., 2021). Additionally, these effects highlight how numerical information is processed spatially and contextually, affecting how we make numerical judgements.

Multi-digit displays

In ecological situations, digits are rarely found in isolation. Indeed, numbers are compositional entities that can be formed by multiple digits, which are horizontally concatenated into strings to express larger quantities. These multidigit displays can be considered as a type of hierarchical stimulus, in which individual digits convey local numerical values, while their spatial arrangement within the string determines a global magnitude that influences numerical judgment. Processing times for number strings are longer than for single numbers, both because of their more complex encoding and their more effortful decoding in terms of the learned place-value system. Nevertheless, the distance effect replicates with two-digit numbers (Zhou et al., 2008; Yan et al., 2024), indicating holistic processing of the entire stimulus string. The same generalization holds for the SNARC effect (Dehaene et al., 1993, Experiment 2), while the size effect might not generalize to double digits (Lavidor et al., 2004). Interestingly, slower responses tend to exhibit stronger SNARC effects than fast responses, indicating a gradual activation of spatial associations over time (Gevers et al., 2006). Overall, this work implies a generalisation of the mental number line model to double-digit numbers.

Another finding from double-digit number comparisons (e.g., which is larger between 35 and 46?) is that people automatically perform separate comparisons of decade and unit numbers, as indicated by faster decisions when these two comparisons align (here: 4 > 3, 6 > 5) than when they conflict (*unit-decade compatibility effect;* Nuerk et al., 2001). This finding indicates decomposition of compound number stimuli, thus violating the holistic processing assumption underlying the mental number line view. However, the compatibility effect depends on familiarity with multi-digit strings and their interpretation; for example, it is strategically modulated, emphasising decade over unit comparisons (reviewed in Nuerk et al., 2015). It is also stronger in women compared to men, and stronger in dense compared to sparse displays (Pletzer et al., 2019). Accounting for such effects of layout, strategies, gender and skill has led to considerably more complex cognitive models of number processing, perhaps requiring a decomposition process, selective attending, and weighting of number components according to task and participant.

A comprehensive model on multi-digit magnitude processing has been formulated by Huber and colleagues (2016; precursors in Moeller et al., 2011; Huber et al., 2013). According to this computational model, the magnitude of each multi-digit stimulus (e.g., tens and units) is represented by a node and processed separately. The comparison networks for single-digit numbers (i.e., tens and units) interact at the level of response selection. Importantly, cognitive control and top-down attention modulate the comparison networks based on digit position and task demands (e.g., by increasing or decreasing the weight assigned to ten or unit nodes). Thus, the Huber et al. (2016) model provides support for a componential over holistic processing of multi-digit numbers, where first a comparison of each single-digit component is performed, followed by a comparison of the multi-digit number.

Although not typically discussed in these terms, the processing of multidigit displays– where local digit recognition is influenced by global spatial arrangement to represent a larger magnitude– offers a window into how our mind deals with hierarchical displays involving several digits at the time. To further explore the role of selective attention in multidigit displays, we now turn to studies where irrelevant digits are presented alongside task-relevant digits.

Selective attention and digits

In some multi-digit tasks, participants report the number of digits on one side of a display while ignoring digits on the other side. Schwarz and Heinze (1998) found that responses were slower and accuracy lower when irrelevant digits were present, regardless of defining characteristics such as color or size. Cohen Kadosh et al. (2007) further demonstrated that irrelevant digits influence the processing of task-relevant digits, highlighting the automatic engagement of numerical cognition processes regardless of task instructions. Both studies with symbolic digits and with non-symbolic dot patterns show that increased irrelevant numerosity of stimuli adversely affects performance, but the effect is more substantial in dot-based tasks due to the visual complexity and spatial distribution of dots compared to symbolic digits.

The impact of selective attention on reaction times (RTs) is significant when subjects must attend to one digit defined by attributes such as color, size, or other dimensions. Increased irrelevant numerical information generally slows responses. Schwarz and Heinze (1998) already showed that the mere presence of irrelevant digits increases cognitive load, resulting in slower processing times. This indicates that participants process both relevant and irrelevant digits to some extent, implying that numerical cognition is highly sensitive to all numerical stimuli within the visual field.

This phenomenon can be explained through the concept of stimuli competing for cognitive resources. Cohen Kadosh et al. (2007) discussed how both relevant and irrelevant stimuli are processed simultaneously, leading to a race for output between them. This competition results in longer RTs as the task-irrelevant digits interfere with processing the relevant ones. This suggests that the brain's mechanisms for digit processing are not entirely selective and that all numerical information is processed to a degree.

Additional studies have examined the influence of task-irrelevant factors, such as the size of digits or the presence of other digits of different sizes, on selective attention. For example, Henik and Tzelgov (1982) found that the physical size of digits can interfere with numerical processing, as larger digits are processed faster than smaller ones, even when size is irrelevant to the task. This supports the notion that task-irrelevant stimuli, such as digit size, are compulsorily processed and can affect the performance on numerical tasks. Several subsequent studies employed the size congruity paradigm or one of several numerical Stroop paradigms, suggesting both an interference of irrelevant non-symbolic magnitude properties (e.g., physical size where symbol meaning and its font size conflict, or numerosity where the number of presented symbols and their meaning conflict) on symbolic number processing, and vice versa (Hershman et al., 2024; Kaufmann et al., 2005; see also Cohen Kadosh et al., 2011 and Hershman et al., 2022 for more details on the nature of the numerical Stroop interference). Such automatic engagement of numerical cognition processes highlights the brain's inherent tendency to process quantitative information holistically, posing challenges for tasks that require selective attention and indicating a fundamental aspect of how numerical information is represented and processed in the brain.

Navon displays

An important question for the validity of cognitive models of number processing is whether performance signatures (i.e., size, distance and SNARC effects) are robust and will be obtained also with multi-digit combinations other than overlearned horizontal number strings. This is the focus of the present study. We investigated this question by presenting hierarchically organised numbers, inspired by the classical Navon letters (Navon, 1977), where a single global digit is composed of a series of smaller local digits, whose identify can be either congruent or incongruent with the global digit.

This seminal study, called "Forest before trees", required participants to classify hierarchical stimuli, consisting of a global and a local level: Large letters made from smaller letters, either congruently (a large 'H' made of small 'H' letters or a large 'S' made of small 'S' letters) or incongruently (a large 'H' made of small 'S' letters or a large 'S' made of small 'H' letters). Importantly, the letters H and S required different key responses to measure processing conflicts, reflected in slower classification times. Navon (1977) manipulated participants' attention by telling them to classify either the global or the local level of randomly presented congruent and incongruent stimuli in separate, counterbalanced trial blocks. He found asymmetrical interference between the global and local levels: Large letters (global level) were quickly classified irrespective of the small ones, while smaller letters (local level) were classified slower when larger ones required the opposite response. This pattern was interpreted as *global precedence*, reflecting the prior entry of larger compared to smaller stimuli into the cognitive processing chain, which leads to faster activation of the response associated with the global letter identity. While Navon (1977) originally assumed this as a serial process (i.e., compulsory processing of global features and later, optional, processing of local features), in a later work he suggested the possibility of parallel processing, with global features being processed faster and, thus, being cognitively available earlier than the local ones (Navon, 1981).

Global precedence has been extensively studied with letters and geometric patterns but only rarely with hierarchical numbers. We briefly review the few studies available to motivate the present experiment. In some studies hierarchical number stimuli were used to study entirely different cognitive mechanisms (Della Libera & Chelazzi, 2006: effects of reward on attention; Mukherjee et al., 2018: pro-sociality following global vs. local priming; Large & McMullen, 2006: conceptual processing following global vs. local priming; Srinivasan & Gupta, 2011: facial emotion processing). Other studies manipulated hierarchical numbers systematically to assess global precedence. For example, Hübner et al. (2001) tested between 8 and 12 participants with hierarchical numbers from 2 to 9 to measure the time cost of attention shifts between global and local levels. Testing 12 young adults, Slavin et al. (2002) replicated the global precedence effect with hierarchical numbers 1 to 4, while older adults and Alzheimer patients showed local precedence. Also using numbers from 1 to 4, Van Leeuwen et al. (2012) reported a reduction of global precedence in 8 meditators.

Since none of the above studies reported numerical cognition signatures, of particular interest here are those few studies that investigated numerical cognition with hierarchical number stimuli. Rosen et al. (2016) replicated the global precedence effect with hierarchical number displays forming arithmetic problems at either the global or local level. However, only a poster presentation is available regarding this study. Bonato and Zorzi (2004, available only in Italian) investigated the automaticity of the SNARC effect by using hierarchical stimuli composed of numbers and oriented triangles. Differently from our present study, number stimuli were only present at one level at the time, while the other level was always composed of triangle shapes. In Experiment 1, participants responded to the parity status of numbers that were presented randomly either at local or global level. A significant SNARC effect emerged both for global and local numbers. In Experiment 2, participants responded to the orientation of the triangles, either at local or global level, while numerical information was completely irrelevant. No evidence of a SNARC effect was found in this experiment, suggesting that the SNARC effect only emerges when the numerical information is task-relevant. Similar results were also found by Prpic et al. (2023) who asked participants to judge the magnitude status of either symbolic or non-symbolic numerals presented as dice patterns. In this study, the two numerical formats were displayed simultaneously, with symbolic numbers always present at the local level and non-symbolic numbers at the global level. Results showed a consistent SNARC effect for symbolic numbers but no effect for non-symbolic numbers. Responses were slower when the magnitude status of the two numerical information differed, indicating that irrelevant numerals were always processed; however, a SNARC effect only emerged when symbolic numbers were task-relevant.

In summary, hierarchical number stimuli offer a complementary but as of yet unused opportunity to study several aspects of numerical processing, such as their encoding, their representation and their processing. This idea is supported by recent work of Pletzer et al. (2021) who tested 91 adults with both letter-based hierarchical stimuli and two-digit number comparisons. The authors reported that global precedence and unit-decade compatibility effects were negatively correlated, indicating that each person applies their individually preferred holistic or analytic processing style to these two tasks. More generally, combining numbers hierarchically can potentially reveal processing constraints for numbers related to the size, distance, and SNARC effect.

Current study

The aim of the current study is to investigate irrelevant number processing at various hierarchical levels in a numerical 'Navon-like' display. In particular, we focused on numerical performance signatures (i.e., size, distance and SNARC effects) to test their robustness and the role of selective attention in modulating the interference effect of irrelevant digits. To do so, in this study, we displayed single digit numbers at both global and local levels and asked participants to classify numerical magnitude (smaller or larger than 5) using two lateralized response keys. Participants were randomly allocated to two groups with different instructions: the global group responded to the global number while ignoring local features, and the local group responded to the local number while ignoring global features. Unlike previous studies, symbolic numbers were simultaneously present at both levels, with attention focus manipulated by instruction. To our knowledge, this is the first study to systematically investigate the size, distance, and SNARC effects using hierarchical number stimuli.

Our stimuli established three experimental conditions within each group (global and local, see Fig. 1b). In the 'Identical' condition, the same digit appeared at both global and local levels (e.g., 1–1). In the 'Response congruent' condition, different digits appeared at global and local levels, but both required the same response (e.g., 4 − 1, both classified as smaller than 5). In the 'Response incongruent' condition, different digits appeared at global and local levels, but they required opposite responses (e.g., 9 − 1, one is classified as larger while the other as smaller than 5).

Based on previous studies on global precedence, we hypothesized that processing global features would be compulsory, with local features being somewhat ignored when attending to the global stimulus, resulting in faster responses in the global group. We also expected processing speeds to vary by condition: fastest in the 'Identical' condition and slowest in the 'Response incongruent' condition. This hypothesis is derived from the Huber et al. (2016) model, which predicts that the 'Identical' condition would activate the same node for both features, while the 'Response congruent' condition would activate separate but non-competing nodes, providing an advantage over the 'Response incongruent' condition.

We further hypothesized that differences in processing speed across conditions would be more pronounced in the local group due to global precedence, with global features being less impacted by local features. Since multiple evidence suggests that number magnitude is automatically processed (Fias et al., 2001; Mitchell et al., 2012; Roth et al., 2025) it is possible that numerical signatures will be elicited simultaneously both for target and irrelevant numbers. Therefore, numerical signatures might compete in the 'Response incongruent' condition, resulting in a null or reduced effect, in particular when irrelevant numbers are at the global level. Conversely, 'Identical' or 'Response congruent' conditions might lead to consistent or even enhanced effects. Alternatively, it is possible that numerical signatures elicited by target numbers will not be moderated by irrelevant digits. Regarding specific numerical signatures, we predicted a larger SNARC effect in the local group due to responses to local numbers being slower than to global numbers, based on the model of Gevers et al. (2006). Regarding the size and distance effects, current literature does not allow us to make any specific prediction.

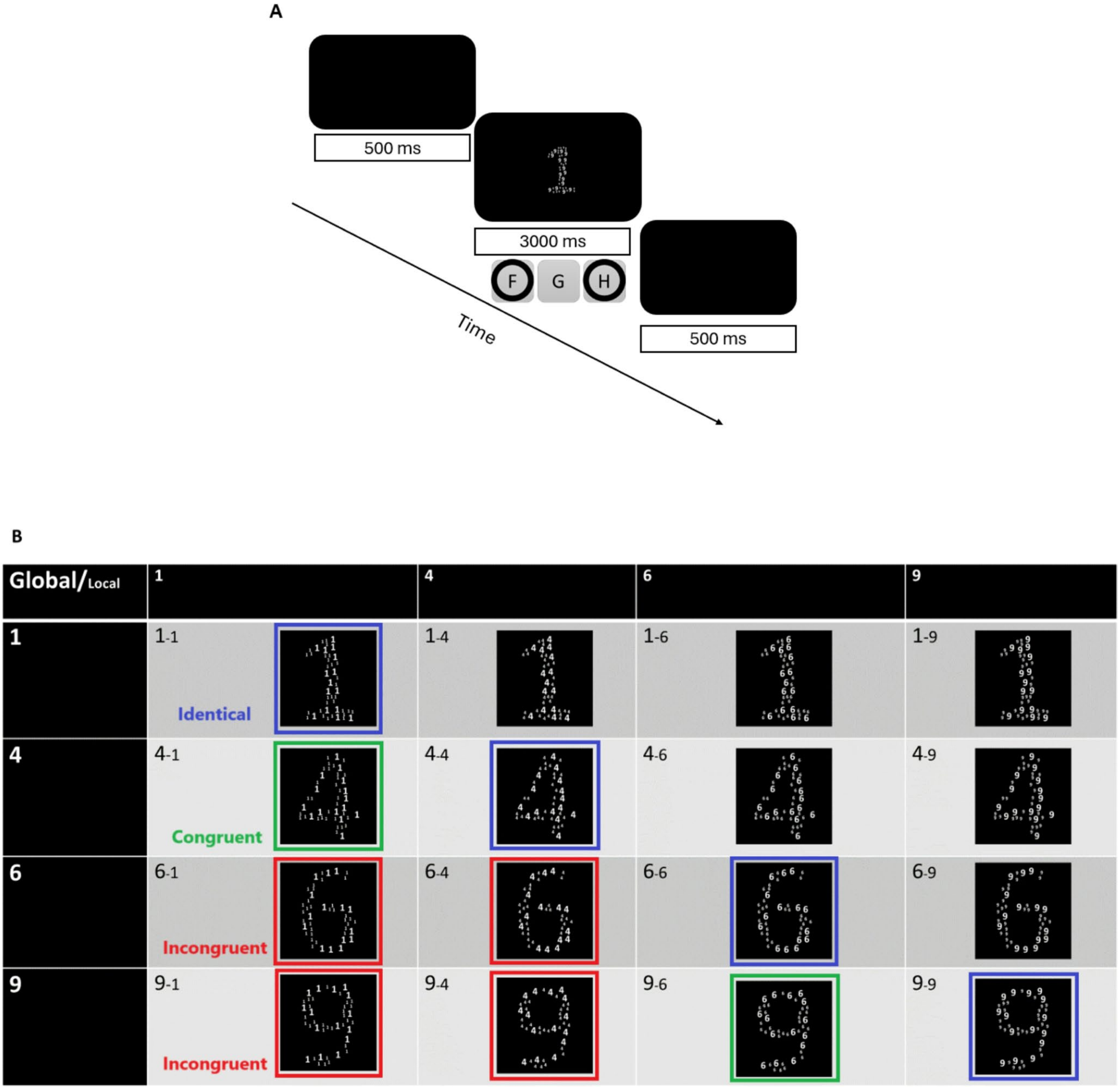

Fig. 1 The panel A displays the sequence of events, with the keyboard buttons "F" and "H". Panel B shows the stimuli with experimental conditions highlighted ('Identical' condition', in **blue**, same digit at both global and local levels (e.g., 1–1); 'Response Congruent' condition, in **green**, different digits at global and local levels, same response (e.g., 4−1, both classified as smaller than 5); 'Response Incongruent' condition, in **red**, different digits at global and local levels, opposite responses (e.g., 9−1, classified as larger and smaller than 5, respectively)

Method

Participants

Sixty-three students (14 male and 49 female) from the University of Bologna participated in the online study. Fifty-eight participants were right-handed and five were left-handed, while the mean age was 22.3 (SD = 2.6). The sample size was calculated by using MorePower 6.4 entering the following parameters: power = 0.90, α = 0.05, $\eta_p^2 = 0.27$ (estimated effect size from the seminal study of Dehaene et al., 1993). The total suggested sample size was 32 (16 participants per group). Nonetheless, previous literature suggests that testing at least 20 participants per group provides sufficient sensitivity to detect the SNARC effect, as well as group differences (Cipora & Wood, 2017). In this Monte Carlo simulation based on the SNARC effect, a sample of 20 participants was considered 'typical' while a sample of 30 participants was considered 'large'. Since our data were collected online, we decided to test at least 30 participants per group in order to compensate for additional noise in measuring response time data (Chetverikov & Upravitelev, 2016). We conservatively estimated our sample size based on the SNARC effect because the literature shows that the distance and the size effect can be reliably detected with a slightly smaller sample (15–20 participants per group; Kojouharova, 2019; additionally, Kochari, 2019 detected a distance effect with 23 participants in web-based experiments).

All participants reported normal or corrected-to-normal vision and exclusively read and wrote in a left-to-right direction. Similarly, the preferred counting direction of all participants was left-to-right, as shown by the dot counting task (Shaki & Fischer, 2023). Written informed consent was obtained from all participants. The experiment was conducted in accordance with the ethical standards of the Declaration of Helsinki and with the approval of the University of Bologna Ethics Committee (Prot. 162247).

Apparatus

The study was designed and controlled through the software for online experiments, *Gorilla Experiment Builder* (Anwyl-Irvine et al., 2020; https://gorilla.sc/). Participants performed the experiments at home on their own device or in the computer rooms of the University of Bologna. Recent studies successfully replicated the SNARC and Distance effects with different stimuli and in different cultures using the Gorilla platform (Felisatti et al., 2024; Fischer et al., 2024).

Stimuli

We term the stimuli used in this study as Hierarchical Digit Navon figures - composite visual patterns designed to assess global and local processing– modelled after the classic Navon stimuli used to investigate hierarchical processing with letters. Each Hierarchical Digit Navon figure consisted of a large digit (global number) composed of a series of smaller digits (local numbers). Both the global and local digits were selected from the set 1, 4, 6 or 9. The global digit formed the overarching shape of the figure, while the local digits filled this shape, repeating to create the composite figure (see Fig. 1B). Stimuli used in the familiarization phase were identical, but generated with either an F or an H as local and global elements. Stimuli were generated as 640×640 pixel images.

Although the experiment was conducted online, the display size of the images was adjusted for each unique display after a calibration procedure (see below). This ensured that each Hierarchical Digit Navon figure subtended approximately 4.1 (vertically) and 2.5 (horizontally) degrees of visual angle, measured from the participant's viewpoint. The size of the global digit was approximately 4.33 cm in height and 2.66 cm in width. The local digits, which comprised the global digit, consisted of different two size elements, which were each approximately 0.36 cm x 0.22 cm and 0.18×0.11 cm for the larger and smaller local digits respectively. Therefore, the larger and smaller local digits were 12 or 24 times smaller than the global digit.

Local digits consisted of two different sizes, generated at 48 or 24 pixels, with a 2:1 ratio between the larger (15) and smaller (30) local elements. This resulted in the global number being determined by the positions of 45 local numbers. This enabled the density and spacing of the stimuli to be kept more uniform at the local level, while still ensuring good visibility of a coherent global number. Both elements had the same identity (i.e. digit 4), ensuring regardless of which local element (of which size) the viewer focused upon, they could resolve the local numerosity task.

During generation of stimuli the local digits were placed sequentially and randomly in order to fill the outline of the global digit. In the event that all elements could not be placed without overlap, the location assignment algorithm reset with new random positions. This procedure was performed ahead of time, ensuring no delays during the experiment itself, and resulted in 10 versions of each digit combination during the experiment. The stimuli were presented in high contrast to maximize visibility. The digits were rendered in white (RGB: 255, 255, 255) against a black background (RGB: 000, 000, 000).

Procedure

At the beginning of the experiment, participants were randomly allocated to two groups so that during the whole study they would only have to focus on either the global or the local dimension of our hierarchical stimuli. After reading the information sheet and providing informed consent, participants were asked to complete the experiment in a quiet room without environmental distractions. To calibrate the stimuli size, thus compensating differences in participant's displays, participants were asked to regulate the size of a credit card image appearing on the screen in order to match that of a regular credit card. After this, participants were asked to familiarise themselves with hierarchical NAVON stimuli (e.g., a global F composed of local H and a global H composed of local F). Participants were asked to respond either to the global or to the local component of our practice stimuli while ignoring the irrelevant dimension. The instructions required to press either the H or the F key, based on stimulus identity and focus instructions (see Fig. 1A). Both speed and accuracy were stressed in the instructions. Participants performed 8 familiarisation trials before each experimental block and received feedback on their performance (correct/incorrect).

After the familiarisation phase, participants started the experimental phase. Participants were informed that in this phase numerical stimuli were employed and that they would have to respond either to the global or local dimension, in accordance with their group allocation. The task was magnitude classification, thus participants were asked to judge whether the target number was smaller or larger than five, while ignoring the irrelevant numbers appearing in the other dimension. The stimuli were presented centrally on the screen and remained visible until a response was made or for a maximum duration of 3000 ms. An inter-trial interval of 500 ms, consisting of a blank screen, followed each stimulus presentation. Participants responded with the F (left) and H (right) keys of their keyboard and they were instructed to keep their index fingers on the response keys throughout data collection. Response time (time from stimulus onset to button press in ms) was recorded, as well as response button identity. Speed and accuracy were again stressed in the instructions and visual feedback (an 'X' over the stimulus for 500 ms) was provided following wrong responses. The experimental phase was divided into four blocks in which the key assignment was manipulated. In two non-consecutive blocks responses to small/large numbers were given with the left/right key, while in the other two non-consecutive blocks key assignment was reversed. Therefore, participants had to change key assignments three times across four blocks during the experiment and were allowed to take a break between blocks. The order of key assignments was counterbalanced among participants. In each block participants completed 160 trials, which consisted of 10 repetitions for each stimulus. This is in line with Cipora and Wood (2017), who suggested to set the number of repetitions to 20 for each key assignment in order to assure sufficient sensitivity to reliably detect the SNARC effect. Based on our conditions, 80 trials per block were the Identical (40 trials) and Response Congruent (40 trials) conditions (in both subcategories global and local digits required the same response, thus they were both smaller or larger than 5), while 80 trials were the Response Incongruent condition (global and local digits required different responses, thus one digit was smaller and one was larger than 5). After completing all four blocks, participants completed a dot counting task (Shaki & Fischer, 2023) and a short version of the Edinburgh handedness inventory (Veale, 2014).

Data analyses and results

Response Times (RTs) of incorrect trials were excluded from the analysis (5.1% in the Global group; 4.2% in the Local group). Similarly, outliers (RTs shorter than 200 ms and those that differed by more than 3 SD from a participant's mean RT) were also removed from data analysis (1.7% in the Global group; 1.6% in the Local group). One participant from each group was excluded from further analyses because they had more than 20% of the total trials removed. We analysed the remaining data using IBM SPSS Statistics software, version 29.0.2.0 (20).

As a first step, we conducted a mixed-factors analysis of variance (ANOVA) on RTs with Group (Local vs. Global) as between participant independent variable (IV) and Condition (Identical, Response Congruent, Response Incongruent) as within participant IV. A significant main effect for Group was found [$F(1, 59) = 17.461$; $p < 0.001$; $\eta^2_p = 0.228$], reflecting significantly faster responses for the Global group (M = 550 ms; SD = 86 ms) compared to the Local group (M = 646 ms; SD = 93 ms). A significant main effect for Condition was also found [$F(2, 118) = 76.255$; $p < 0.001$; $\eta^2_p = 0.564$]. Pairwise comparisons (Bonferroni adjusted) indicated significantly faster RTs for the Identical (M = 582 ms; SD = 97 ms) compared to the Response Congruent (M = 603 ms; SD = 103 ms; $p < 0.001$) and Incongruent (M = 606 ms; SD = 104 ms; $p < 0.001$) conditions. Conversely, the Response Congruent and Incongruent conditions did not differ significantly ($p = 0.679$). Furthermore, a significant interaction emerged between Group and Condition [$F(2, 118) = 27.48$; $p < 0.001$; $\eta^2_p = 0.32$]. Pairwise comparisons (Bonferroni adjusted) were performed as follows.

In the Global group, the Identical condition (M = 544 ms; SD = 84 ms) differed significantly from the Response Congruent (M = 555 ms; SD = 86 ms; $p = 0.002$) and Incongruent (M = 552 ms; SD = 89 ms; $p = 0.027$) conditions, conversely the Response Congruent and Incongruent conditions did not differ ($p = 0.904$). Since responses in the Identical condition were significantly faster than in the other two conditions, this suggests that irrelevant digits were processed at the local level, and this interfered with the processing of the target digit at the global level.

In the Local group, the Identical condition (M = 622 ms; SD = 95 ms) differed significantly from the Response Congruent (M = 654 ms; SD = 95 ms; $p < 0.001$) and Incongruent (M = 661 ms; SD = 89 ms; $p < 0.001$) conditions. Furthermore, in the Local group also the Response Congruent and Incongruent conditions differed significantly ($p = 0.024$). These results suggest that irrelevant digits were processed at the global level and interfered with the processing of target digits at the local level. Since the Response Congruent

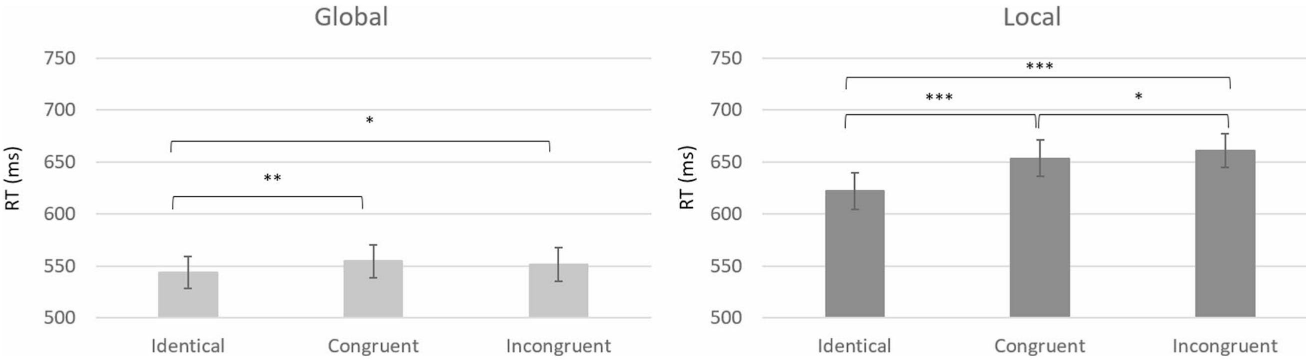

Fig. 2 Average response times (RT) for the Global (left panel) and Local (right panel) groups across experimental conditions (Identical, Response Congruent and Response Incongruent). Significance levels are indicated as follows: * indicates significance at $p<0.05$, ** indicates significance at $p<0.01$, *** indicates significance at $p<0.001$. Error bars indicate standard error of the mean

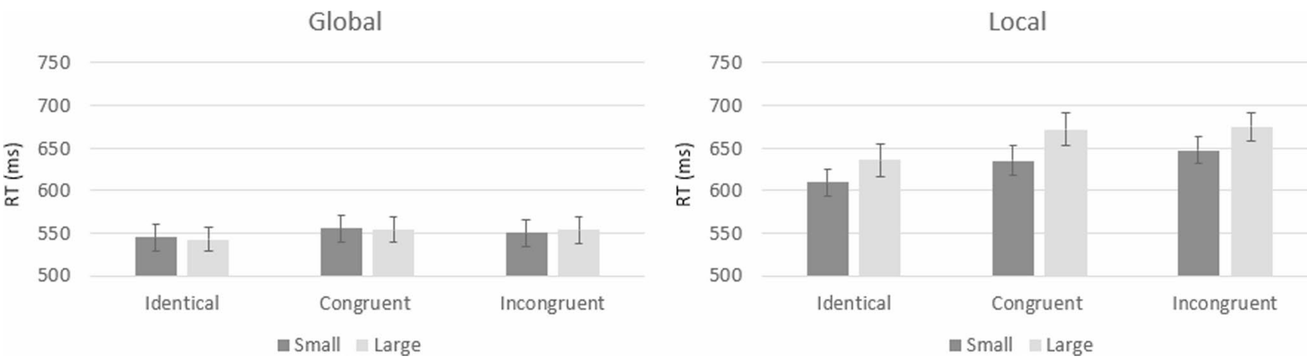

Fig. 3 Size effect in the Global (left panel) and Local (right panel) groups. Average response times (RT) for numerical size (Small and Large) are reported across experimental conditions (Identical, Response Congruent and Response Incongruent). Error bars indicate standard error of the mean

condition was faster than the Response Incongruent condition, it seems that not only the identity of the stimuli but also the magnitude of the irrelevant global numbers was processed. That is, although in both conditions the identity of digits differed between local and global levels, in the Response Congruent condition both digits were classified as either smaller or larger than 5, leading to a response advantage compared to the Response Incongruent condition Fig. 2.

As a second step, we conducted three different mixed design ANOVAs to compare the size, SNARC and distance effects across the Global and Local groups.

Size effect

We performed a mixed design ANOVA on RTs with Group (Global vs. Local) as a between participant IV, Size (Small: numbers 1 and 4; vs. Large: numbers 6 and 9) and Condition (Identical, Response Congruent, Response Incongruent) as within participants IVs. A significant main effect for Size [$F(1, 59)=19.604$; $p<0.001$; $\eta_p^2 = 0.249$] was found, as well as a significant interaction between Size and Group [$F(1, 59) = 19.136$; $p < 0.001$; $\eta_p^2 = 0.245$]. Pairwise comparisons (Bonferroni adjusted), performed for the interaction between Group and Size, showed that a size effect was present in the Local ($p<0.001$) but not in the Global ($p = 0.97$) group. Indeed, in the Local group, small target numbers were processed faster (M = 631 ms; SE = 16 ms) than large numbers (M=661; SE=17 ms), while this was not true for the Global group (Small numbers M = 550, SE=16 ms; Large numbers M=550, SE=16 ms). Conversely, Size did not interact with Condition ($p = 0.353$) nor with Group and Condition ($p = 0.215$). This suggests that the size effect in the Local group was not moderated by Condition Fig. 3.

SNARC effect

We performed a mixed design ANOVA on dRTs (RTs of right– left key) with Group (Global vs. Local) as a between participant IV, Size (Small: numbers 1 and 4; vs. Large: numbers 6 and 9) and Condition (Identical, Response Congruent, Response Incongruent) as within participants IVs. Positive values indicate an advantage for left key responses, while negative values indicate an advantage for right key responses. The only significant main effect was Size [$F(1, 59)=11.593$; $p=0.001$; $\eta_p^2 = 0.164$], confirming the presence of a SNARC effect. This resulted in a left key advantage for small numbers (M = 10 ms; SE = 5 ms) and a

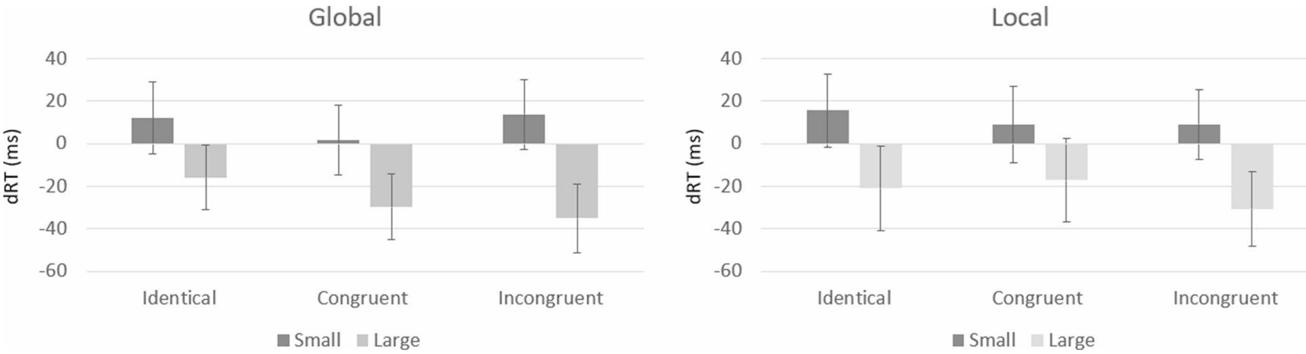

Fig. 4 SNARC effect in the Global (left panel) and Local (right panel) groups. Average difference in response times (dRT = right key– left key) for numerical size (Small and Large) are reported across experimental conditions (Identical, Response Congruent and Response Incongruent). Error bars indicate standard error of the mean

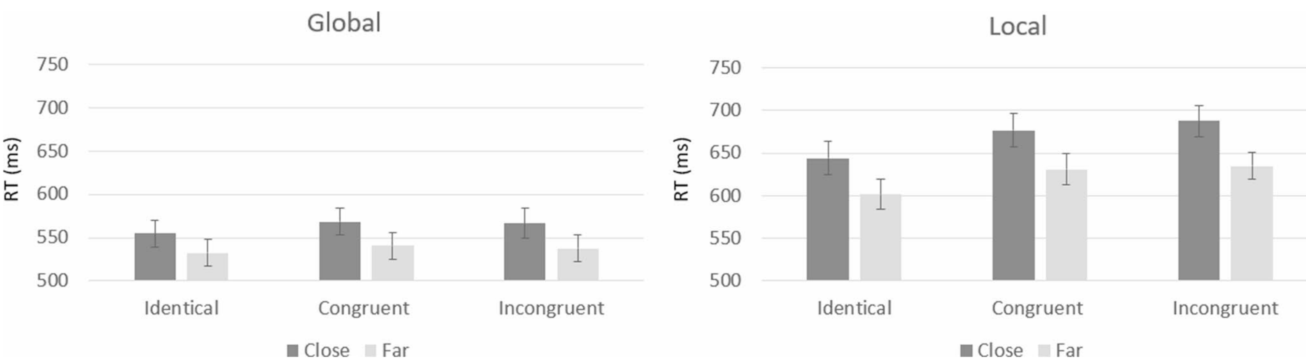

Fig. 5 Distance effect in the Global (left panel) and Local (right panel) groups. Average response times for numerical distance (Close and Far) are reported across experimental conditions (Identical, Response Congruent and Response Incongruent). Error bars indicate standard error of the mean

right key advantage for large numbers (M = −25 ms; SE = 7 ms). Conversely, Size did not interact with Group (p = 0.932), Condition (p = 0.166) nor with Group and Condition (p = 0.563). This suggests that the SNARC effect was not moderated by Group or by Condition Fig. 4.

Distance effect

We performed a mixed design ANOVA on RTs with Group (Global vs. Local) as between participant IV, Distance (Close to reference: numbers 4 and 6; vs. Far from reference: numbers 1 and 9) and Condition (Identical, Response Congruent, Response Incongruent) as within participants IVs. A significant main effect for Distance [$F(1, 59) = 192.782$; $p < 0.001$; $\eta^2_p = 0.766$] was found (Close M = 616 ms, SE = 12 ms, Far M = 580 ms, SE = 11 ms), as well as a significant interaction between Distance and Group [$F(1, 59) = 15.027$; $p < 0.001$; $\eta^2_p = 0.203$]. Pairwise comparisons (Bonferroni adjusted), performed for the interaction between Distance and Group, showed that a distance effect was present both in the Local (‘Close’ M = 670 ms, SE = 17 ms; ‘Far’ M = 623 ms, SE = 16 ms; $p < 0.001$) and the Global (‘Close’ M = 563 ms, SE = 17 ms; ‘Far’ M = 537 ms, SE = 16 ms; $p < 0.001$) group. Conversely, Distance did not interact with Condition (p = 0.139) nor with Group and Condition (p = 0.776). This suggests that the distance effect was not moderated by Condition. Furthermore, results confirm the presence of a distance effect in both groups, although this effect appears to be larger in the Local (Close-Far = 47 ms) compared to the Global (Close-Far = 26 ms) group Fig. 5.

General discussion

In this study, we employed ‘Navon-like’ hierarchical number stimuli– termed Hierarchical Digit Navon figures– to investigate various aspects of numerical processing. Numbers (1, 4, 6 and 9) were simultaneously present both at global and local levels while participants were instructed to classify target numbers as smaller or larger than 5. Participants were divided into two separate groups: a global group (respond to the global and ignore local number) and a local group (respond to local and ignore the global number). As expected, according to global precedence, results clearly show that stimuli were processed faster at the global level. Furthermore, both in the global and local group, target stimuli were identified faster when the same number was present both at global and local levels. Conversely, only in the local group a response advantage was found for the Response congruent conditions (local and global numbers differ but are both small or large in magnitude) compared to the Response incongruent conditions (local and global numbers differ also in magnitude). This supports our first hypothesis, namely that processing of the global stimulus would be compulsory while local numbers could be ignored to some degree.

In accordance with conclusions by Navon (1981), who opened up to the possibility of parallel processing, our evidence shows that local features can interfere with the processing of the global stimulus. Indeed, as mentioned earlier, a global number was processed faster when it shared the same identity with irrelevant local numbers. However, no processing advantage was found in the global group when the global number differed from the irrelevant local numbers but both were classified as small or as large (i.e., Response congruent condition). Conversely, when participants responded to the local stimuli, a response advantage was found in the Response congruent condition too. This is again in line with global precedence, which predicts a larger interference of global features during local processing, than vice versa. Accordingly, our evidence also suggests that the impact of the irrelevant global number on local processing was more consistent than the impact of local numbers on global processing. Indeed, in the local group, the processing advantage of the identical condition over the congruent and incongruent conditions was about three times larger (in terms of ms) than in the global group.

As suggested by Navon (1977, 1981), global features are processed faster and, thus, available earlier than local ones. Thus, it is plausible that number identity information (same vs. different), which is computed faster than the magnitude (small vs. large) of different numbers, can interfere also during global processing. Conversely, information about numerical magnitude, which is computed slower, can interfere only during local processing.

Our findings are also in line with the Huber et al. (2016) model of multi-digit magnitude processing, supporting its claims with evidence from non-horizontal multi-digit displays. According to this model, identical numbers will activate the same node, leading to a processing advantage for the 'Identical condition' also during global processing. Conversely, when two numbers differ, they activate two separate nodes. However, if these numbers are both classified as small or large, they will not compete at the response selection stage, thus providing a smaller yet detectable response advantage during the processing of local features. Overall, our findings are also in line with previous literature on selective attention which suggests that irrelevant numbers are automatically processed independently from task instructions (Cohen Kadosh et al., 2007; Schwarz & Heinze, 1998) is present in the text, but reference is missing in the reference section. Could you please provide the reference or delete the text citation." Resolved="yes"

A key goal of this study was to investigate whether mandatory processing of numbers would lead both irrelevant and target digits to simultaneously elicit numerical signatures (size effect, SNARC effect and distance effect), and whether these would interfere to some degree. This query was motivated by evidence suggesting that number magnitude is automatically processed (Fias et al., 2001; Mitchell et al., 2012; Roth et al., 2025). Based on this, we hypothesised that in the Response incongruent condition numerical signatures might compete (e.g., target number is small but irrelevant number is large). Consequently, either absent or reduced effects might be detected due to interference at the representational or response level. Following this logic, it was possible to expect consistent or even increased effects in the Response congruent and Identical conditions. Furthermore, such interference between numerical signatures might be moderated by the processing level (global/local) of our hierarchical stimuli. Indeed, when the target is at the global level, less interference could arise from the competing local stimuli, while larger interference could arise from the irrelevant global number in the local group. Finally, although there is no support for more specific hypotheses regarding the size and distance effects, a computational model predicted a larger SNARC effect with slower responses (Gevers et al., 2006). According to this model, a larger SNARC effect was predicted in the Response incongruent conditions and in the local group.

The present results provide mixed support for these predictions. In the global group both a significant SNARC and distance effect emerged, while there was no evidence of a size effect. Conversely, in the local group all numerical signatures (size, SNARC and distance effects) were found. Interestingly, all numerical signatures were consistent across condition levels, suggesting that these were not prone to interference from irrelevant number features. Thus, differently from our expectations, numerical signatures did not compete or interact, remaining stable across different conditions. Comparing numerical signatures across groups confirmed the presence of the size effect only in the local group and showed that the distance effect was also stronger in this group. This might be due to longer response times in the local group, which could foster stronger response compatibility effects. However, longer response times did not impact the SNARC effect, which remained quite stable across groups, differently from what could be expected from the model of Gevers and colleagues (2006).

Our findings are overall consistent with the limited evidence from previous studies that investigated the SNARC effect with hierarchical stimuli. For example, Bonato and

Zorzi (2004) found that, when participants judged the parity status of target numbers, a similar SNARC effect was found both at local and global levels. This supports our finding that the SNARC effect does not change significantly when numbers are present as local stimuli or as a global figure. In their second experiment, Bonato and Zorzi (2004) asked participants to judge the orientation of triangles while numbers were task irrelevant. No SNARC effect was found, either at local or global level, suggesting that irrelevant numbers do not interfere with response selection when the attentional focus is on a different hierarchical level (e.g., target = local, irrelevant number = global). This might be the case also in our study and would explain why the magnitude of irrelevant numbers did not interfere with the SNARC effect elicited by the target stimulus.

Our results are also in line with the findings of Prpic et al. (2023). In their study, multiple numbers were displayed to form dice-like patterns and participants were asked to respond either to symbolic numerals (digits/local) or to non-symbolic quantities (numerosity/global). Similar to our study, they found a response advantage when identical/congruent numerals were present both at local and global levels, supporting the idea that irrelevant numbers can be processed automatically and independently from the processing level. Nevertheless, the magnitude of irrelevant stimuli did not moderate the SNARC effect found for symbolic numerals, which remained consistent, as in our study. Conversely, no SNARC effect emerged for numerosity. Overall, evidence suggests that irrelevant numbers did not moderate the response patterns elicited by target numbers.

Despite these similarities, several differences can be found between the two studies. For example, Prpic et al. (2023) did not differentiate between identical and magnitude congruent conditions, thus it is not possible to conclude if number identity is processed faster than number magnitude, as our study suggests. Furthermore, in Prpic et al. (2023) symbolic numbers were always displayed at the local level, while non-symbolic numerals were displayed at the global level. Therefore numerical format (symbolic/non-symbolic) and level of processing (local/global) were confounded. Although this confound did not allow to draw clear conclusions, our present study suggests that a SNARC effect can be elicited by numerical stimuli that require global processing. Thus, the different outcome for symbolic and non-symbolic numerals in Prpic et al. (2023) is more likely due to numerical format rather than to levels of processing.

Local and global image components can be conveyed by different ranges of *spatial frequencies* (SFs; Sergent, 1982; Lamb & Yund, 1993). SFs are low-level visual features corresponding to the number of dark-light cycles per degree of visual angle. Perceptual systems decompose visual stimuli into their underlying spatial frequency content. Different ranges of SFs convey different information: Low SF range (few cycles per degree of visual angle) represents coarse shape; high SF range (many cycles per degree of visual angle) corresponds to fine details. From the re-integration of different SFs results the identification of a stimulus. Remarkably, behavioral, neuropsychological, and neuroimaging evidence has shown that the SFs processing is lateralized in the brain (review in Roberston & Ivry, 2000; Felisatti et al., 2020a): The left hemisphere preferentially processes fine signals, while the right hemisphere is superior in filtering coarse signals. Recently, Felisatti and colleagues (2020a,2020b, 2020c) proposed the *Brain's Asymmetric Frequency Tuning* (BAFT) model which explains spatial biases in behavior by integrating the asymmetry in SF selection with the asymmetry of attentional orienting (Kinsbourne, 1970). BAFT entails implications for different cognitive domains (Felisatti et al., 2020a) and especially for numerical cognition (Felisatti et al., 2020b, c). Indeed, according to BAFT, the early processing of low/high SFs with the right/left hemisphere and the perception of coarse/fine features in the left/right visual field would lead to extrapolation of the regularity between less/more and the left/right space, respectively. An interesting hypothesis to test is the role of the brain's asymmetric frequency tuning in spatial-numerical association with symbolic numbers. The Hierarchical Digit Navon stimuli used in the present study represent an ideal test material for this hypothesis.

The current study is the first attempt to investigate numerical signatures by using Hierarchical Digit Navon stimuli, which present different digits simultaneously both at global and local levels. Our findings thus provide novel evidence and should be generalized to other hierarchical numerical stimuli. In particular, the degree of interference between global and local features could be manipulated by making each stimulus level more or less salient (e.g., by manipulating the size or density of the stimuli). Therefore, as the global stimulus becomes more salient we should expect a stronger global precedence and, possibly, larger interference on numerical signatures elicited by local stimuli. Future studies should further manipulate the salience of global and local features and investigate how this could interfere with numerical processing. Furthermore, in our experiment we employed a magnitude classification task which is characterised by explicit magnitude processing and therefore it is likely to elicit a stronger effect for the target stimulus (e.g., SNARC effect, Macnamara et al., 2018; Wood et al., 2008). Indeed, it is possible that more subtle effects due to the processing of irrelevant features (e.g., orientation or colour; Fias et al., 2001; Roth et al., 2025) would be more prone to interference from irrelevant digits. Therefore, future studies should employ a variety of tasks in order to systematically investigate the interplay between local and global numbers in hierarchical displays.

By manipulating task relevance and examining how participants respond to numerically relevant versus irrelevant stimuli at these different scales, our Hierarchical Digit Navon displays enable a nuanced understanding of how numerical cognition is influenced by spatial attention and hierarchical processing. This approach not only sheds light on the interaction between task-relevant and irrelevant numerical stimuli but also enhances our understanding of how cognitive resources are allocated in numerosity tasks. More specifically, this interaction is crucial for understanding the dynamics of selective attention and cognitive control in these tasks. Additionally, the findings contribute to our understanding of the underlying mechanisms of numerical processing, highlighting how these mechanisms are interconnected with broader cognitive functions such as working memory, attention, and executive control. These insights not only inform numerical cognition but also provide broader implications for understanding how individuals manage cognitive resources in complex tasks.

In conclusion, in this study we investigated how numbers are processed in a context of hierarchical stimuli by presenting numbers both at local and global level. We found that numbers at the global level were responded faster, as expected by global precedence, and that the number identity of irrelevant numbers was always processed, independently from the focus level. Furthermore, when responding to local numbers the magnitude of irrelevant global stimuli was extracted and impacted response times. Conversely, this was not true when responding to global numbers, suggesting that local features could be ignored at some degree. Regarding numerical signatures, the SNARC effect appeared to be unaffected by the level of processing, while the size effect only appeared for local target numbers and the distance effect seemed also more pronounced at the local level. Finally, numerical signatures remained stable across condition levels, suggesting that irrelevant numbers did not interfere with the size, SNARC and distance effects elicited by target numbers.

Author contributions All authors contributed to the study conception and design. DAM prepared the stimuli, AF programmed the online experiment and VP analysed the data. All authors contributed to writing the first draft and commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Data availability Data is provided within the manuscript or supplementary information files.

Declarations

Competing interests The authors declare no competing interests.

References

Anwyl-Irvine, A. L., Massonnié, J., Flitton, A., et al. (2020). Gorilla in our midst: An online behavioral experiment builder. *Behavior Research Methods, 52*, 388–407. https://doi.org/10.3758/s13428-019-01237-x

Bonato, M., & Zorzi, M. (2004). È automatico attivare Una rappresentazione spaziale Ogni Volta Che Vediamo dei numeri? *Giornale Italiano Di Psicologia*, *31*(3), 615–622. https://doi.org/10.1421/16266

Chetverikov, A., & Upravitelev, P. (2016). Online versus offline: The web as a medium for response time data collection. *Behavior Research Methods*, *48*, 1086–1099. https://doi.org/10.3758/s13428-015-0632-x

Cipora, K., & Wood, G. (2017). Finding the SNARC instead of hunting it: A 20∗ 20 Monte Carlo investigation. *Frontiers in Psychology*, *8*, 243273. https://doi.org/10.3389/fpsyg.2017.01194

Cohen Kadosh, R., Gevers, W., & Notebaert, W. (2011). Sequential analysis of the numerical Stroop effect reveals response suppression. *Journal of Experimental Psychology: Learning Memory and Cognition*, *37*(5), 1243. https://doi.org/10.1037/a0023550

Dehaene, S., & Cohen, L. (1995). Toward an Anatomical and Functional Model of Number Processing. *Mathematical Cognition, 1*, 83–120.

Dehaene, S., Bossini, S., & Giraux, P. (1993). The mental representation of parity and number magnitude. *Journal of Experimental Psychology: General*, *122*(3), 371. https://doi.org/10.1037/0096-3445.122.3.371

Felisatti, A., Aagten-Murphy, D., Laubrock, J., Shaki, S., & Fischer, M. H. (2020a). The brain's asymmetric frequency tuning: Asymmetric behavior originates from asymmetric perception. *Symmetry*, *12*(12), 2083. https://doi.org/10.3390/sym12122083

Felisatti, A., Laubrock, J., Shaki, S., & Fischer, M. H. (2020b). A biological foundation for spatial–numerical associations: The brain's asymmetric frequency tuning. *Annals of the New York Academy of Sciences*, *1477*(1), 44–53. https://doi.org/10.1111/nyas.14418

Felisatti, A., Laubrock, J., Shaki, S., & Fischer, M. H. (2020c). Commentary: A mental number line in human newborns. *Frontiers in Human Neuroscience*, *14*, 99. https://doi.org/10.3389/fnhum.2020.00099

Felisatti, A., Ranzini, M., Shaki, S., & Fischer, M. H. (2024). Distance in depth: A comparison of explicit and implicit numerical distances in the horizontal and radial dimensions. *Journal of Experimental Psychology: General*. https://doi.org/10.1037/xge0001537

Fias, W., Lauwereyns, J., & Lammertyn, J. (2001). Irrelevant digits affect feature-based attention depending on the overlap of neural circuits. *Cognitive Brain Research*, *12*(3), 415–423. https://doi.org/10.1016/S0926-6410(01)00078-7

Fischer, M. H., Felisatti, A., Li, X., & Shaki, S. (2024). A cross-cultural comparison of finger-based and symbolic number representations. *Journal of Experimental Child Psychology*, *246*, 105979. https://doi.org/10.1016/j.jecp.2024.105979

Gevers, W., Verguts, T., Reynvoet, B., Caessens, B., & Fias, W. (2006). Numbers and space: A computational model of the SNARC effect. *Journal of Experimental Psychology: Human Perception and Performance*, *32*(1), 32. https://doi.org/10.1037/0096-1523.32.1.32

Henik, A., & Tzelgov, J. (1982). Is three greater than five: The relation between physical and semantic size in comparison tasks. *Memory & Cognition*, *10*, 389–395. https://doi.org/10.3758/BF03202431

Hershman, R., Beckmann, L., & Henik, A. (2022). Task and information conflicts in the numerical Stroop task. *Psychophysiology*, *59*(9), e14057. https://doi.org/10.1111/psyp.14057

Hershman, R., Beckmann, L., Keha, E., Wagner, M., Kaufmann, L., & Henik, A. (2024). A color-digit Stroop task shows numerical influence on numerosity processing. *Memory & Cognition*, 1–12. https://doi.org/10.3758/s13421-024-01631-7

Huber, S., Moeller, K., Nuerk, H. C., & Willmes, K. (2013). A computational modeling approach on three-digit number processing. *Topics in Cognitive Science*, *5*(2), 317–334. https://doi.org/10.1111/tops.12016

Huber, S., Nuerk, H. C., Willmes, K., & Moeller, K. (2016). A general model framework for multisymbol number comparison. *Psychological Review*, *123*(6), 667. https://doi.org/10.1037/rev0000040

Hübner, R., Futterer, T., & Steinhauser, M. (2001). On attentional control as a source of residual shift costs: Evidence from two-component task shifts. *Journal of Experimental Psychology: Learning Memory and Cognition*, *27*(3), 640. https://doi.org/10.1037/0278-7393.27.3.640

Kadosh, R. C., Kadosh, K. C., Linden, D. E., Gevers, W., Berger, A., & Henik, A. (2007). The brain locus of interaction between number and size: A combined functional magnetic resonance imaging and event-related potential study. *Journal of Cognitive Neuroscience*, *19*(6), 957–970. https://doi.org/10.1162/jocn.2007.19.6.957

Kaufmann, L., Koppelstaetter, F., Delazer, M., Siedentopf, C., Rhomberg, P., Golaszewski, S., & Ischebeck, A. (2005). Neural correlates of distance And congruity effects in a numerical Stroop task: An event-related fMRI study. *Neuroimage*, *25*(3), 888–898. https://doi.org/10.1016/j.neuroimage.2004.12.041

Kinsbourne, M. (1970). The cerebral basis of lateral asymmetries in attention. *Acta Psychologica*, *33*, 193–201. https://doi.org/10.1016/0001-6918(70)90132-0

Kochari, A. R. (2019). Conducting web-based experiments for numerical cognition research. *Journal of Cognition*, *2*(1). https://doi.org/10.5334/joc.85

Kojouharova, P. S. (2019). The numerical distance and size effects in symbolic numerical cognition. Doctoral dissertation. Eötvös Loránd University, Budapest. https://www.ppk.elte.hu/dstore/document/184/kojouharova_petia_disszertacio.pdf

Lamb, M. R., & Yund, E. W. (1993). The role of Spatial frequency in the processing of hierarchically organized stimuli. *Perception & Psychophysics*, *54*(6), 773–784. https://doi.org/10.3758/BF03211802

Large, M. E., & McMullen, P. A. (2006). Hierarchical attention in discriminating objects at different levels of specificity. *Perception & Psychophysics*, *68*, 845–860. https://doi.org/10.3758/BF03193706

Lavidor, M., Brinksman, V., & Göbel, S. M. (2004). Hemispheric asymmetry and the mental number line: Comparison of double-digit numbers. *Neuropsychologia*, *42*(14), 1927–1933. https://doi.org/10.1016/j.neuropsychologia.2004.05.003

Leeuwen, S. V., Singer, W., & Melloni, L. (2012). Meditation increases the depth of information processing. *Frontiers in Human Neuroscience*, 6(133). https://doi.org/10.3389/fnhum.2012.00133

Libera, C. D., & Chelazzi, L. (2006). Visual selective attention and the effects of monetary rewards. *Psychological Science*, *17*(3), 222–227. https://doi.org/10.1111/j.1467-9280.2006.01689.x

Macnamara, A., Keage, H. A., & Loetscher, T. (2018). Mapping of non-numerical domains on space: A systematic review and meta-analysis. *Experimental Brain Research*, *236*, 335–346. https://doi.org/10.1007/s00221-017-5154-6

Mitchell, T., Bull, R., & Cleland, A. A. (2012). Implicit response-irrelevant number information triggers the SNARC effect: Evidence using a neural overlap paradigm. *Quarterly Journal of Experimental Psychology*, *65*(10), 1945–1961. https://doi.org/10.1080/17470218.2012.673631

Moeller, K., Huber, S., Nuerk, H. C., & Willmes, K. (2011). Two-digit number processing: Holistic, decomposed or hybrid? A computational modelling approach. *Psychological Research Psychologische Forschung*, *75*, 290–306. https://doi.org/10.1007/s00426-010-0307-2

Moyer, R. S., & Landauer, T. K. (1967). Time required for judgements of numerical inequality. *Nature*, *215*(5109), 1519–1520. https://doi.org/10.1038/2151519a0

Mukherjee, S., Srinivasan, N., Kumar, N., & Manjaly, J. A. (2018). Perceptual broadening leads to more prosociality. *Frontiers in Psychology*, *9*, 1821. https://doi.org/10.3389/fpsyg.2018.01821

Navon, D. (1977). Forest before trees: The precedence of global features in visual perception. *Cognitive Psychology*, *9*(3), 353–383. https://doi.org/10.1016/0010-0285(77)90012-3

Navon, D. (1981). The forest revisited: More on global precedence. *Psychological Research Psychologische Forschung*, *43*(1), 1–32. https://doi.org/10.1007/BF00309635

Nuerk, H. C., Moeller, K., Klein, E., Willmes, K., & Fischer, M. H. (2015). Extending the mental number line. *Zeitschrift Für Psychologie*. https://doi.org/10.1027/2151-2604/a000041

Nuerk, H. C., Weger, U., & Willmes, K. (2001). Decade breaks in the mental number line? Putting the tens and units back in different bins. *Cognition*, *82*(1), B25–B33. https://doi.org/10.1016/S0010-0277(01)00142-1

Pitt, B., & Casasanto, D. (2022). The order of magnitude: Why SNARC-like tasks (Still) cannot support a generalized magnitude system. *Cognitive Science*, *46*(2), e13108. https://doi.org/10.1111/cogs.13108

Pletzer, B., Harris, T., & Scheuringer, A. (2019). Sex differences in number magnitude processing strategies are mediated by Spatial navigation strategies: Evidence from the unit-decade compatibility effect. *Frontiers in Psychology*, *10*, 229. https://doi.org/10.3389/fpsyg.2019.00229

Pletzer, B., Scheuringer, A., Harris, T., & Scherndl, T. (2021). The missing link: Global-local processing relates to number-magnitude processing in women. *Journal of Experimental Psychology: General*, *150*(3), 560. https://doi.org/10.1037/xge0000951

Prpic, V., Basamh, Y. A., Goodridge, C. M., Agostini, T., & Murgia, M. (2023). Contrasting symbolic and non-symbolic numerical representations in a joint classification task. *Psychonomic Bulletin & Review*, *30*(4), 1422–1430. https://doi.org/10.3758/s13423-023-02246-w

Prpic, V., Mingolo, S., Agostini, T., & Murgia, M. (2021). Magnitude and order are both relevant in SNARC and SNARC-like effects: A commentary on Casasanto and Pitt (2019). *Cognitive Science*, *45*(7), e13006. https://doi.org/10.1111/cogs.13006

Robertson, L. C., & Ivry, R. (2000). Hemispheric asymmetries: Attention to visual and auditory primitives. *Current Directions in Psychological Science*, *9*(2), 59–63. https://doi.org/10.1111/1467-8721.00061

Rosen, M., Mills, M., & Dodd, M. (2016). Task-Defined requirements of attention and Global-Local processing. *Journal of Vision*, *16*(12), 693–693. https://doi.org/10.1167/16.12.693

Roth, L., Caffier, J., Reips, U. D., Cipora, K., Braun, L., & Nuerk, H. C. (2025). True colors SNARC: Semantic number processing is highly automatic. *Journal of Experimental Psychology: Learning Memory and Cognition*. https://doi.org/10.1037/xlm0001431

Schwarz, W., & Heinze, H. J. (1998). On the interaction of numerical and size information indigit comparison: A behavioral and event-related potential study. *Neuropsychologia, 36*(11), 1167–1179. https://doi.org/10.1016/S0028-3932(98)00001-3

Sergent, J. (1982). The cerebral balance of power: Confrontation or cooperation? *Journal of Experimental Psychology: Human Perception and Performance*, *8*(2), 253. https://doi.org/10.1037/0096-1523.8.2.253

Shaki, S., & Fischer, M. H. (2023). How do numbers shift Spatial attention? Both processing depth and counting habits matter. *Journal of Experimental Psychology: General*. https://doi.org/10.1037/xge0001493

Sixtus, E., Krause, F., Lindemann, O., & Fischer, M. H. (2023). A sensorimotor perspective on numerical cognition. *Trends in Cognitive Sciences*, *27*(4), 367–378. https://doi.org/10.1016/j.tics.2023.01.002

Slavin, M. J., Mattingley, J. B., Bradshaw, J. L., & Storey, E. (2002). Local–global processing in alzheimer's disease: An examination of interference, Inhibition And priming. *Neuropsychologia*, *40*(8), 1173–1186. https://doi.org/10.1016/S0028-3932(01)00225-1

Srinivasan, N., & Gupta, R. (2011). Rapid communication: Global–local processing affects recognition of distractor emotional faces. *Quarterly Journal of Experimental Psychology*, *64*(3), 425–433. https://doi.org/10.1080/17470218.2011.552981

Veale, J. F. (2014). Edinburgh handedness inventory–short form: A revised version based on confirmatory factor analysis. *Laterality: Asymmetries of Body Brain and Cognition*, *19*(2), 164–177. https://doi.org/10.1080/1357650X.2013.783045

Wood, G., Willmes, K., Nuerk, H. C., & Fischer, M. H. (2008). On the cognitive link between space and number: A meta-analysis of the SNARC effect. *Psychology Science Quarterly*, *50*(4), 489.

Yan, D., Vijayakumar, S., & Hong, J. (2024). EXPRESS: The effects of psychological distance on the diagnosticity of digits to the left versus right of a separator. *Journal of Marketing Research*. https://doi.org/10.1177/00222437241262799

Zhou, X., Chen, C., Chen, L., & Dong, Q. (2008). Holistic or compositional representation of two-digit numbers? Evidence from the distance, magnitude, and SNARC effects in a number-matching task. *Cognition*, *106*(3), 1525–1536. https://doi.org/10.1016/j.cognition.2007.06.003